



ba·bash·ka

Babashka: fast native Clojure scripting

Clojure/Conj 2026

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`(prn :Hello :world!)`

`@borkdude | @lispyclouds`

Agenda

Part 1 ~1h

- What and Why of **bb** (~2m)
- Editor and REPL setup (~8m)
- Setting up the project (~10m)
- A tour of the land (~20m)
- A spooky, cryptic detour (~10m)
- Behind the Scenes and References (~10m)
- A little break (~10m)

Agenda

Part 2 ~2h

- Can we build it? (~110m)
 - Exercise 1: Using and testing scrypt (10m)
 - Exercise 2: Wiring up clipboard support (10m)
 - Exercise 3: Implement local storage of site meta (15m)
 - Exercise 4: Wiring up the CLI and Spectre v3 logic (26m)
 - Exercise 5: Completing the Charm TUI (28m)
 - Exercise 6: Publishing, installing and using bbin (15m)



What and Why of bb

IT DOES HAVE A COOL NAME RIGHT?

A close-up photograph of a white ceramic coffee cup filled with a latte, featuring a heart-shaped latte art design. The cup sits on a matching saucer on a dark wooden table. The background is blurred, showing what appears to be a window with light coming through.

Clojure on the JVM rocks, but...

- We've been the butt of many jokes about the **slow start-up time**. The slow start-up time is for good reasons.
- **Too much ceremony** just to write a simple script.
- Damn, the python/bash/ruby **scripts run quick!**
- The clojure+java ecosystem is great but **one day I can parse JSON without a dependency!**
- lein, deps, tools.build etc etc... all **I want is a simple thing to automate** my boring tasks.
- Yeah, might as well bite the bullet and write bash/python/make. **Good that I have ChatGPT too along with StackOverflow.**



What if you could keep writing Clojure?

KEEP THE NICE LISPY MUSCLE MEMORY GOING

Who is bb for?

WHERE ALL CAN IT BE USED?

- The ~~biased~~ experienced Clojure dev **can keep writing their scripts in Clojure.**
- Folks needing more **scripting/glue code** to automate parts of their code and deployments.
- Folks working closer to the infra and **not on the JVM can still write Clojure.** Can this finally kill the stupid YAML?
- Running Clojure on **resource constrained environments as cross-platform and self contained scripts.**
- **Need for (start-up) speed:** Write quick and lean Clojure scripts for functions as a service like AWS Lambda.
- **Sneak in Clojure as more maintainable scripts** in non Clojure codebases or teams.



Scripting issues

NOT A BED OF ROSES

- FACT: **No one can write shell scripts** correctly.
- FACT: **No one can write Makefiles** correctly. Also, *WHAT* is up with the **.PHONY** tabs?
- The shell scripts are **not cross platform**. Windows shouldn't be ignored.
- **Complexity creeps up** in the automation code fast.
- **Big mental shift** when writing scripts.
- Pack everything in docker? **Ship your machine?**
- Would be nice to not only **keep writing Clojure** here but also **have the same reuse and maintainability** as the app code.





Editor and REPL setup

BALANCE THE ((()))

bb --version

clojure --version

Let the editor wars begin!

The indispensable Clojure LSP: <https://clojure-lsp.io/installation/>

Neovim

<https://github.com/Olical/conjure>

Emacs

<https://docs.cider.mx/cider/index.html>

VSCode/Codium

<https://calva.io/>

Vim

<https://github.com/liquidz/elin> (written in bb!)

Sublime Text 4

<https://github.com/tonsky/Clojure-Sublimed>

IntelliJ IDEA

<https://cursive-ide.com/>

script.clj

```
(defn fact
  [n]
  (reduce * 1 (range 1 (inc n))))

;; do the pythonistas here recognise this? ;)
(when (= *file* (System/getProperty "babashka.file"))
  (println (fact 5)))

(comment
  (fact 5))
```



Setting up the project

FINALLY SOME CODE!

The source code

```
git clone https://github.com/babashka/babashka-workshop-spectre
```

Trying it out

- Running tests: **bb test**
- Try connecting your editor and try out eval-ing



A tour of the land





babashka.{fs,process,ffi,cli}



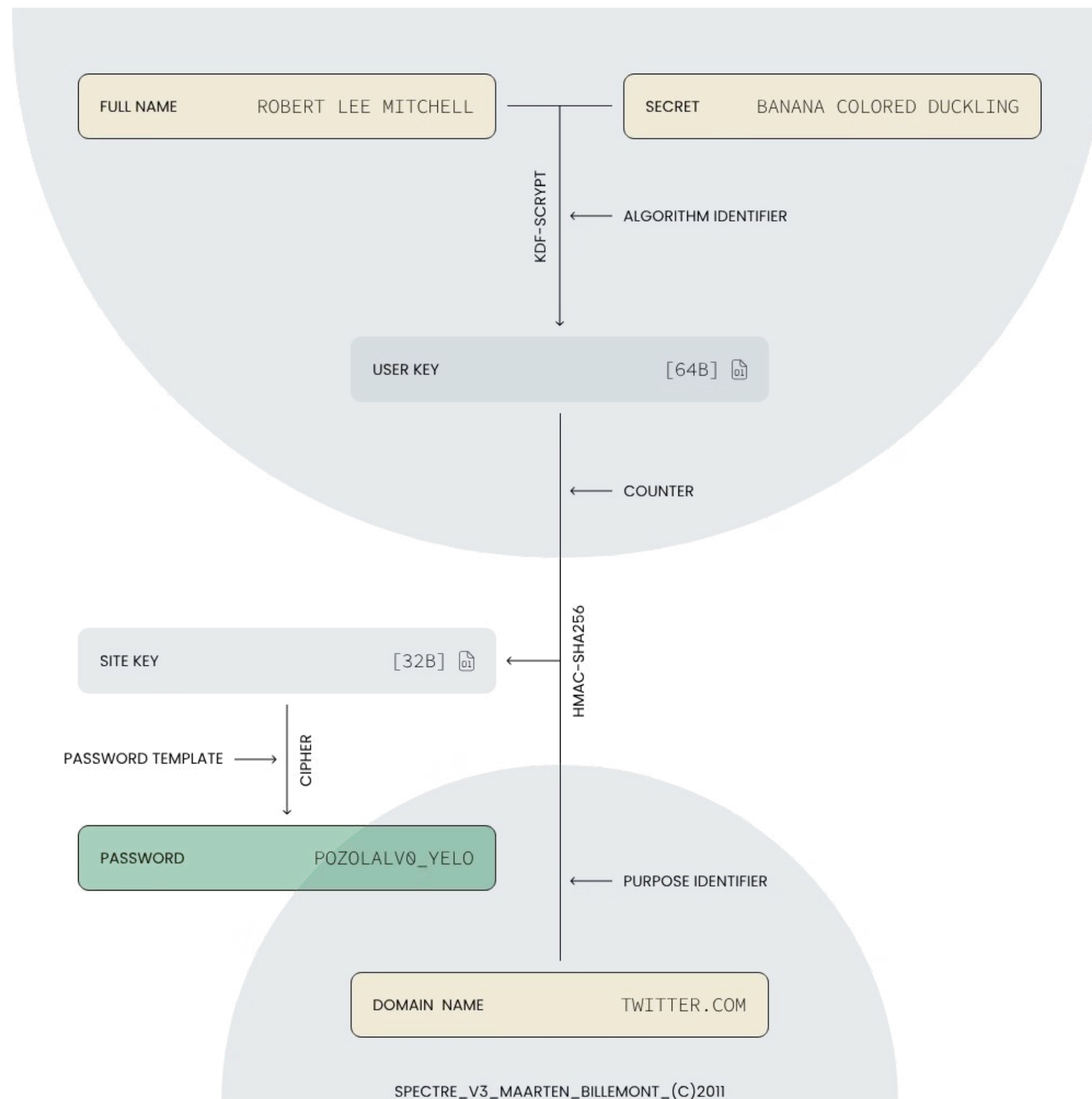
bb.edn and tasks

RIP MAKE (MAYBE WE SHOULD'VE CALLED IT THAT)

- <https://book.babashka.org/#project-setup>
- <https://book.babashka.org/#tasks>



A spooky, cryptic detour





The algorithm

<https://spectre.app/>

Phase 1: The identity

```
masterKey = SCRIPT ( key, seed, N, r, p, dkLen )
```

```
key = <master password>
```

```
seed = scope . LEN(<name>) . <name>
```

```
N = 32768
```

```
r = 8
```

```
p = 2
```

```
dkLen = 64
```

Phase 2: The site key

```
siteKey = HMAC-SHA-256 ( key, seed )
```

```
key = <master key>
```

```
seed = scope . LEN(<site name>) . <site name> . <counter>
```

Phase 3: The site password

```
template = templates [ <site key>[0] % LEN( templates ) ]
```

```
for i in 0..LEN( template ):
```

```
    passChars = templateChars [ template[i] ]
```

```
    passWord[i] = passChars[ <site key>[i+1] % LEN( passChars ) ]
```



Behind the Scenes

THE REAL POWER OF BABASHKA

The Dude



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- <https://ko-fi.com/borkdude>

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>Nextjournal

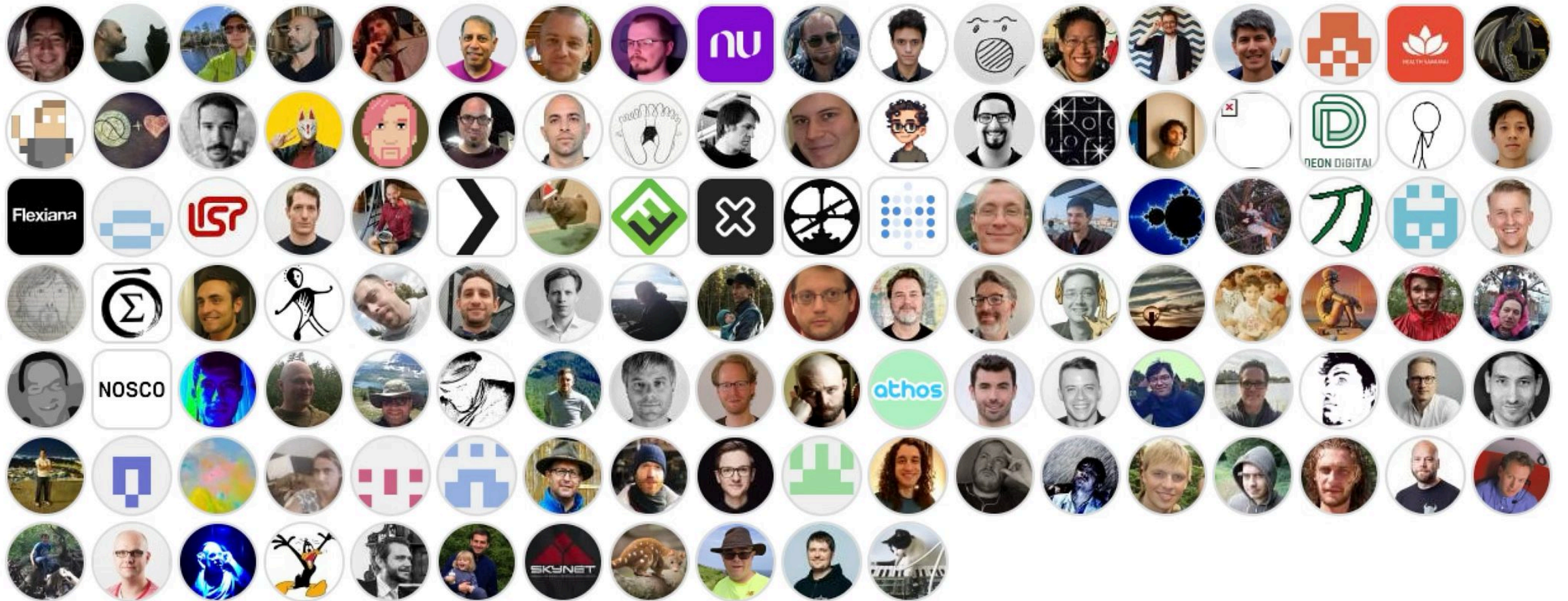
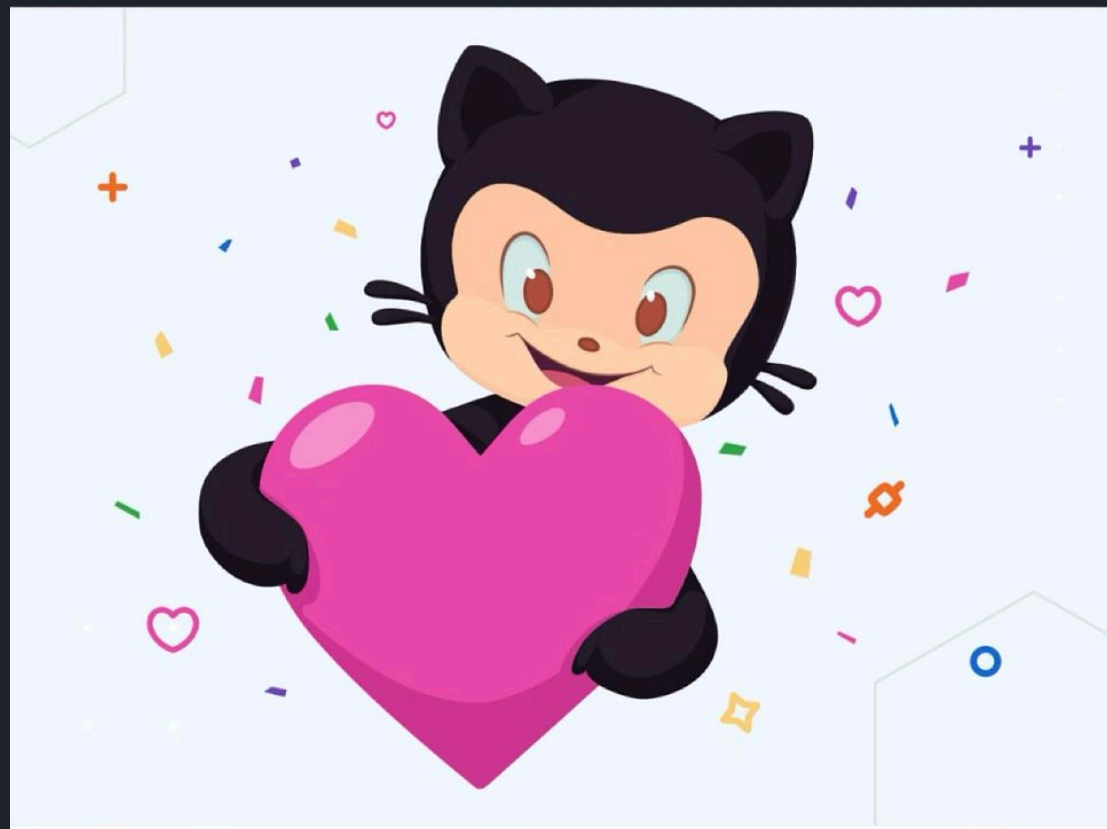
Roam Research



clojurists
together

Nubank

nu



References

ONE-STOP SHOP

- Site: <https://babashka.org>
- The Book: <https://book.babashka.org>
- The Babooka: <https://www.braveclojure.com/quests/babooka>
- The wiki: <https://github.com/babashka/babashka/wiki>
- GitHub: <https://github.com/babashka/babashka>
- Clojurians Slack: [#babashka](#)
- Collection of more external resources:
<https://github.com/babashka/babashka#articles-podcasts-and-videos>
- The spectre algorithm: <https://spectre.app/>
- Follow the **#babashka** hashtag and the Babashka [News](#).





Contributing

- Code lives on: <https://github.com/babashka/babashka>
- Come **chat with us** on the [#babashka](#) channel on the Clojurians Slack.
- Checkout the dev [doc](#) and [build](#) doc.
- Recommended steps:
 - **Chat** before creating an issue on Clojurians Slack
 - **Look for an issue** that may already address your needs.
 - **Raise a Pull Request(PR)** from a fork containing:
 - The code **changes**
 - **Tests** for the changes
 - **Changelog**
 - **Docs** if needed

INTERMISSION
BE BACK SOON



Can we build it?

YES WE CAN!



Thank You!

MAY THE START-UP TIME BE WITH YOU!



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